

Smooth morphisms: extension properties

SGA 1, Exposé III

English draft

A. Grothendieck

About this draft

This is an unofficial English translation of Exposé III, “Morphismes lisses: propriétés de prolongement”, from SGA 1. The source volume is A. Grothendieck and M. Raynaud, *Revêtements étales et groupe fondamental* (SGA 1), Séminaire de géométrie algébrique du Bois Marie, 1960–61. It follows the slightly corrected SMF recomposition, [arXiv:math/0206203v2](https://arxiv.org/abs/math/0206203v2).

This draft contains all seven sections of the exposé, including proofs, footnotes, diagrams, and the final application to formal and ordinary smooth schemes over a complete local ring. Statement numbering and mathematical notation follow the source. References to material outside this exposé retain their source numbers. Apparent source issues have not been silently repaired; scholarly proofreading remains outstanding.

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EXPOSÉ III

Smooth morphisms: extension properties

1. Formally smooth homomorphisms

In II, we limited ourselves to the consideration of homomorphisms of finite type, and consequently, in local homomorphisms $A \rightarrow B$ of local rings, to the case where B is isomorphic to a localization of an A -algebra of finite type. This case is insufficient for various applications, notably in formal geometry or analytic geometry. For example, the formal power-series ring $B = A[[t_1, \dots, t_n]]$ has, from the point of view of formal geometry, the properties of a smooth algebra over A . In analytic geometry, the same is true of the local ring of a point (y, z) of a product $Y \times \mathbf{C}^n$ considered as an algebra over the local ring of y ; moreover, the completion of this algebra is isomorphic to the algebra of formal power series in n indeterminates over the completion of the base ring $\mathcal{O}_x$. This leads to the following definition.

DEFINITION 1.1. Let $u: A \rightarrow B$ be a local homomorphism of local rings (noetherian, as recalled). Suppose $k(B)$ is finite over $k(A)$. We say that u is a *formally smooth homomorphism*, or that the algebra B is *formally smooth over A* , if there exists a local finite $\bar{A}$ -algebra A' , free over $\bar{A}$, such that the local components of the semi-local ring $\bar{B} \otimes_{\bar{A}} A' = B'$ are A' -isomorphic to algebras of formal power series over A'^1 .

(We denote by $\bar{A}, \bar{B}$ the completed rings of A, B .) Since B' is finite and free over $\bar{B}$, it is indeed a semi-local ring, a direct sum of complete local rings, each of which is still a free module over $\bar{B}$, hence has the same dimension as $\bar{B}$, and therefore as B . It follows that the number of variables t_i in the formal power-series rings considered in 1.1 is equal to $\dim \bar{B} - \dim \bar{A} = \dim B - \dim A$, and in particular is independent of the local component chosen. One sees at once that it is also the dimension of the ring $B \otimes k = B/\mathfrak{m}B$, where $k = A/\mathfrak{m}$ is the residue field of A ; we shall call it the *relative dimension of B with respect to A* .

REMARK. It is evident that Definition 1.1 depends only on the homomorphism on the completions $\bar{A} \rightarrow \bar{B}$ induced by $A \rightarrow B$, which justifies the terminology to some extent. We repeat here of Definitions I I.3.2 b) and I I.4.1 b), which risk being misleading, and prefer to say “formally unramified” respectively “formally étale” in the cases considered in these definitions, reserving the terminology “unramified” and “étale” for the case where B is a localization of an A -algebra of finite type². The reader will immediately verify that “formally étale” is equivalent to “formally smooth and quasi-finite”. Finally, let us point out that there is a reasonable definition of “formally smooth” without any prior hypothesis on the residual extension $k(B)/k(A)$ (supposed finite here), encompassing among others the local homomorphisms $A \rightarrow B$ such that B is *flat* over A and $B/\mathfrak{m}B$ is a *separable extension* of $A/\mathfrak{m} = k$ (not necessarily of finite type); for example, a Cohen p -ring is formally *smooth* over the ring of p -adic integers. It is the lifting property for homomorphisms (compare 2.1) that should become the definition in this general case. For the applications we have in view, the case treated in Definition 1.1 will suffice; in what follows, by “formally smooth” we shall understand “with finite residual extension”.

LEMMA 1.2. *If B is formally smooth over A , then B is flat over A .*

¹For a more general and more conceptual definition, motivated by 2.1 below, cf. EGA 0_{IV} 19.3.1.

²Or better, “essentially unramified” respectively “essentially étale”, compare EGA IV 18.6.1.

Since flatness is invariant under completion, we may suppose that A and B are complete. Since flatness is invariant under a local flat (hence faithfully flat) extension of the base ring, Definition 1.1 reduces us to the case where B is a formal power-series algebra over A . But then, as an A -module, B is isomorphic to a product of A -modules isomorphic to A , hence (the base ring A being noetherian) is A -flat as a product of flat A -modules.

Let us place ourselves under the conditions of 1.1. Since the residual extensions of the local components of B' over A' are trivial, it follows that $L \otimes_k k'$ is an artinian k' -algebra whose local components have trivial residual extensions (where L, k, k' are the residue fields of A, B, A'). This necessary condition for the finite free extension A' to satisfy the condition stated in 1.1 is also sufficient, as follows at once from 1.3 (i) and 1.4 below.

PROPOSITION 1.3. *Let $A \rightarrow B$ be a local homomorphism of local rings, with finite residual extension, and let A' be a finite local A -algebra over A , such that $B' = B \otimes_A A'$ is finite over B , hence a semi-local ring, also noetherian. (i) If B is formally smooth over A , then the localizations of B' at its maximal ideals are formally smooth over A' . (ii) The converse is true if A' is free over A .*

We are immediately reduced to the case where A, B are complete.

(i) Let A'' be a finite free local extension of A such that the local components of $B'' = B \otimes_A A''$ are formal power-series algebras over A'' . Extending scalars $A'' \rightarrow A'' \otimes_A A' \rightarrow A'''$, where A''' is a local component of $A'' \otimes_A A'$, one sees that the local components of $B'' \otimes_{A''} A''' = B \otimes_A A'''$ are formal power-series algebras over A''' . But one also has

$$B \otimes_A A''' = (B \otimes_A A') \otimes_{A'} A''' = B' \otimes_{A'} A''',$$

while, since A'' is free over A , $A'' \otimes_A A'$ is free over A' , and consequently so is A''' , which is a direct factor of it. This proves that B' is formally smooth over A' .

(ii) Let A'' be a finite free local A' -algebra such that the local components of $B' \otimes_{A'} A'' = B \otimes_A A''$ are formal power-series algebras over A'' . Since A' is free over A , so is A'' , and hence B is formally smooth over A .

PROPOSITION 1.4. *Let $A \rightarrow B$ be a local homomorphism of local rings, with trivial residual extension. For B to be formally smooth over A , it is necessary and sufficient that $\overline{B}$ be isomorphic to a formal power-series algebra over $\overline{A}$.*

Only the necessity has to be proved, and we may suppose that A and B are complete. Let $\mathfrak{m}$ ($\mathfrak{n}$) be the maximal ideal of A (B), and let $t_1, \dots, t_n$ be elements of $\mathfrak{n}$ defining a basis of the vector space

$$(\mathfrak{n}/\mathfrak{n}^2)/\mathfrak{S}(\mathfrak{m}/\mathfrak{m}^2) = \mathfrak{n}/(\mathfrak{n}^2 + \mathfrak{m}B).$$

These elements therefore define a homomorphism of local A -algebras

$$B_1 = A[[t_1, \dots, t_n]] \longrightarrow B.$$

We prove that it is an isomorphism. It suffices to prove that for every power $\mathfrak{m}^q$ of $\mathfrak{m}$, one obtains an isomorphism after reducing modulo $\mathfrak{m}^q$ (since B_1 and B are the projective limits of the corresponding rings reduced modulo $\mathfrak{m}^q$, with q variable). Since B and B_1 are flat A -modules, the graded rings associated with the $\mathfrak{m}$ -adic filtration are obtained by tensoring over $k = A/\mathfrak{m}$ with $\text{gr}(A)$ the rings $B_1/\mathfrak{m}B_1$ respectively $B/\mathfrak{m}B$. We are thus reduced to showing that $B_1/\mathfrak{m}B_1 \rightarrow B/\mathfrak{m}B$ is an isomorphism. In view of 1.2, we are thereby reduced to the case where A is a field k . On the other hand, if A' is a finite free local A -algebra such that $B \otimes_A A'$ is a formal power-series algebra over A' (note that this algebra is local, since the residual extension of B over A is trivial), to prove that $B_1 \rightarrow B$ is an isomorphism, it suffices to prove that $B_1 \otimes_A A' \rightarrow B \otimes_A A'$ is one. This reduces us to the case where B is already a formal power-series algebra (this reduction should have been made first, before reducing to the case of a base field). But then B is a regular local ring with coefficient field k , and it is well known (and immediate by considering the graded rings associated with the $\mathfrak{n}_1$ -adic and $\mathfrak{n}$ -adic filtrations on B_1 and B) that $B_1 \rightarrow B$ is an isomorphism. This completes the proof.

COROLLARY 1.5. *If B is formally smooth over A , then there exists a finite local A -algebra A' such that the local components of*

$$\overline{B} \otimes_{\overline{A}} \overline{A'} = \overline{(B \otimes_A A')}$$

are isomorphic to formal power-series algebras over $\overline{A'}$.

Indeed, if L/k is the residual extension of B/A , consider an extension k'/k such that the residual extensions in the k' -algebra $L \otimes_k k'$ are trivial. Let A' be an algebra finite and free over A such that $A'/\mathfrak{m}A' = k'$ (one knows that such an algebra exists, for example by reducing step by step to the case where k'/k is monogenic, and then lifting to A the coefficients of the minimal polynomial of a generator of k' over k). It is local. Then $B \otimes_A A'$ has, at its maximal ideals, trivial residual extensions over that k' of A' , and the conclusion follows with the help of 1.4.

COROLLARY 1.6. *Let $A \rightarrow B$ be a local homomorphism of local rings. For B to be formally smooth over A , it is necessary and sufficient that B be flat over A and that $B/\mathfrak{m}B$ be formally smooth over $k = A/\mathfrak{m}$.*

Making a suitable finite free local extension A' of A and using 1.3 (ii), we are reduced to the case where the residual extension of B/A is trivial. We know moreover by 1.3 (i) and 1.2 that the stated conditions are necessary. For the sufficiency, it suffices to observe that the proof of 1.4 proves, under the hypotheses made here, that B is a formal power-series algebra over A (supposing A and B complete, which is permissible).

REMARK. It would not be difficult to develop, for formally smooth homomorphisms, the analogue of all the properties of smooth morphisms studied in II. For the differential properties, however, this requires a modification of the usual definition of Kähler differentials (cf. I 1.1), with completed tensor products replacing ordinary tensor products. We shall content ourselves with evoking these abysses here, what precedes being sufficient for our purpose.

It remains to make the link between formal smoothness and the notion of smoothness developed in II (and of which we have not yet made any use):

PROPOSITION 1.7. *Let $A \rightarrow B$ be a local homomorphism, with B a localization of an A -algebra of finite type. For B to be smooth over A , it is necessary and sufficient that it be formally smooth over A .*

Using 1.6 and II.2.1, we are reduced to the case where A is a field.

Using 1.3 (ii) and II.4.13, a suitable extension k' of k reduces us to the case where the residual extension for B/k is trivial. In view of 1.4 (respectively II.5.2), B is then smooth over k (respectively formally smooth over k) if and only if B is a regular local ring (respectively its completion is a formal power-series algebra over k). But it is well known that these two conditions are equivalent (the residual extension being trivial).

2. The lifting property characteristic of formally smooth homomorphisms

THEOREM 2.1. *Let $A \rightarrow B$ be a local homomorphism of local rings defining a finite residual extension. The following conditions are equivalent:*

- (i) *B is formally smooth over A*
- (ii) *For every local homomorphism $A \rightarrow C$, where C is a local ring complete, every ideal J of C contained in $\mathfrak{r}(C)$, and every local A -homomorphism $B \rightarrow C/J$, there exists an A -homomorphism (necessarily local) $B \rightarrow C$ which lifts it.*
- (iii) *For every A -algebra C (not necessarily a noetherian ring), every nilpotent ideal J of C , and every continuous A -homomorphism $B \rightarrow C$ (i.e. vanishing on a power of $\mathfrak{r}(B)$), there exists an A -homomorphism $B \rightarrow C$ (necessarily continuous as well) which lifts it.*
- (iv) *The same statement as (ii) and (iii), but with C a local artinian ring, finite over A .*

(iv bis) As in (iv), but with J moreover square-zero.

REMARK. For the rest of this exposé, we shall use only the implication (iv) $\Rightarrow$ (i), or (iv bis) $\Rightarrow$ (i); the direct implication (i) $\Rightarrow$ (ii) will be proved by another method in the next no. when B is a localization of an algebra of finite type over A . Let us recall that in the “good” theory of Cohen theorems³, property (ii) or (iii) becomes the definition of formally smooth homomorphisms, whereas 1.1 becomes a characteristic property valid only in the case of a finite residual extension. Care should be taken that of properties (ii) and (iii), neither is more general than the other. One could give an equivalent property covering both, by introducing a linearly topologized ring C , *separated* and *complete*, a *closed topologically nilpotent* ideal of C , and a continuous homomorphism $A \rightarrow C$ (thus making C a topological A -algebra); we shall leave this modification to the reader.

Proof of 2.1. We shall prove (i) $\Rightarrow$ (iii) $\Rightarrow$ (ii), then (iv) $\Rightarrow$ (i). Since (ii) $\Rightarrow$ (iv) is trivial, and the equivalence of (iv) and (iv bis) is seen by an immediate induction on the integer n such that $J^n = 0$, this will finish the proof.

(i) $\Rightarrow$ (iii). An immediate induction reduces us to the case where $J^2 = 0$. Since C is finite over A , there exists a power $\mathfrak{m}^q$ of the maximal ideal of A which annihilates C . Dividing by $\mathfrak{m}^q$, and noting that $B/\mathfrak{m}^q B$ is still formally smooth over $A/\mathfrak{m}^q$ by 1.3 (i), we may suppose that A is artinian. Since B is flat over A by 1.2, B is free over A , since A is artinian. Thus there exists an A -module homomorphism

$$w: B \longrightarrow C$$

which lifts the given homomorphism $u: B \rightarrow C/J$. Put

$$f(x, y) = w(xy) - w(x)w(y) \quad (x, y \in B)$$

then $f(x, y) \in J$, and f is therefore an A -bilinear map from $B \times B$ into J . For there to exist a lift $v: B \rightarrow C$ of u which is an algebra homomorphism, it is necessary and sufficient that there exist an A -linear map $g: B \rightarrow J$ such that $v = w + g$ is an algebra homomorphism, which is written

$$\begin{aligned} g(1) &= 1 - w(1) \\ g(x, y) - u(x)g(y) - u(y)g(x) &= -f(x, y) \quad (\text{for } x, y \in B) \end{aligned}$$

This is a system of *linear* equations in $\text{Hom}_A(B, J)$, with right-hand sides in J ; hence it has a solution if and only if the corresponding system in $\text{Hom}_A(B, J) \otimes_A A'$, with right-hand sides in $J' = J \otimes_A A'$, has a solution, A' denoting a faithfully flat algebra over A . Now let A' be an algebra finite and free over A , local, such that $B' = B \otimes_A A'$ is a formal power-series algebra over A' (N.B. in our proof one may suppose A and B complete, as is immediately seen). Since A' is free of finite type over A , we have

$$\text{Hom}_A(B, J) \otimes_A A' = \text{Hom}_{A'}(B', J')$$

and one verifies that the system of equations obtained in $\text{Hom}_{A'}(B', J')$ is the one which determines the homomorphisms of A' -algebras $B' \rightarrow C' = C \otimes_A A'$ which lift the homomorphism $u': B' \rightarrow C'/J'$ deduced from u by extension of scalars, by “correcting” by an A' -module homomorphism $g': B' \rightarrow J'$ the A' -module homomorphism $w': B' \rightarrow C'$ deduced from w by extension of scalars. (Note that B generates B' as an A' -module.) We are thus reduced to proving (iii) when B is a *formal power-series algebra* over A , $B = A[[t_1, \dots, t_n]]$. Let us lift arbitrarily the images in C/J of the t_i to elements z_i of C . Since the z_i modulo J are nilpotent ($u: B \rightarrow C/J$ being continuous), the z_i themselves are nilpotent (since J is nilpotent), and hence the z_i define a continuous homomorphism of topological A -algebras from B into the discrete ring C , evidently lifting u , qed.

(iii) $\Rightarrow$ (ii). Let $\mathfrak{n}$ be the maximal ideal of C , and for every integer $q > 0$, put

$$C_q = C/\mathfrak{n}^q, \quad J_q = (J + \mathfrak{n}^q)/\mathfrak{n}^q$$

³Cf. EGA 0IV 19.3, 19.8

Thus C_q/J_q identifies with a quotient algebra of C/J . On the other hand, the composite homomorphism $u_q: B \rightarrow C/J \rightarrow C_q/J_q$ is continuous from B into the discrete ring C_q/J_q , and J_q is a nilpotent ideal in C_q . We then construct, step by step, A -homomorphisms

$$v_q: B \longrightarrow C_q$$

such that (a) v_q lifts u_q and (b) v_q lifts v_{q-1} . The possibility of the induction is easily verified, since

$$u_q: B \longrightarrow C/(J + \mathfrak{n}^q) \quad \text{and} \quad v_{q-1}: B \longrightarrow C/\mathfrak{n}^{q-1}$$

define the same homomorphism

$$B \longrightarrow C/((J + \mathfrak{n}^q) + \mathfrak{n}^{q-1}) = C/(J + \mathfrak{n}^{q-1}) = C_{q-1}/J_{q-1}$$

namely u_{q-1} , they define a homomorphism

$$B \longrightarrow C/J'_q \quad \text{where} \quad J'_q = (J + \mathfrak{n}^q) \cap \mathfrak{n}^{q-1} \supset \mathfrak{n}^q$$

(from which both arise by reduction). We are therefore reduced to lifting a homomorphism $B \rightarrow C/J'_q$ from B into a quotient of C_q by an ideal $J'_q/\mathfrak{n}^q$ contained in J_q , hence nilpotent, and this is possible by hypothesis (iii).

This done, the v_q define a homomorphism from B into the projective limit C of the C_q . Since J is closed, J is the projective limit of the J_q ; hence v lifts u , qed.

(iv) $\Rightarrow$ (i). One observes first at once that if (iv) is verified, (iv) remains verified for the local components of $B \otimes_A A'$ over A' , if A' is a finite local algebra over A . Taking A' free over A and such that the residual extensions of B' over A' are trivial, we are reduced, by 1.3 (ii), to the case where the residual extension of B over A is trivial. We shall then prove the following slightly more precise result:

COROLLARY 2.2. *Under the conditions of 2.1, suppose moreover that the residual extension of B over A is trivial. Then the equivalent conditions of 2.1 are also equivalent to the following two conditions (supposing in (v) that A and B are complete):*

(iv ter) *As in (iv), but with the local artinian ring C finite over A restricted to have a trivial residual extension (and moreover, if one wishes, with the ideal J square-zero).*

(v) *There exists a local A -homomorphism (where $n = \dim \mathfrak{n}/(\mathfrak{n}^2 + \mathfrak{m}B)$)*

$$u: B \longrightarrow B_1 = A[[t_1, \dots, t_n]]$$

inducing an isomorphism

$$\mathfrak{n}/(\mathfrak{n}^2 + \mathfrak{m}B) \xrightarrow{\sim} \mathfrak{n}_1/(\mathfrak{n}_1^2 + \mathfrak{m}B_1)$$

where $\mathfrak{n}$ ($\mathfrak{n}_1$) is the maximal ideal of B (B_1), and $\mathfrak{m}$ is that of A .

Proof. Since (iv bis) evidently implies (iv ter)—setting aside the joke about the square-zero ideal—it will suffice to prove (iv ter) $\Rightarrow$ (v) $\Rightarrow$ (i).

(iv ter) $\Rightarrow$ (v). Choose a basis $a_1, \dots, a_n$ of $\mathfrak{n}/(\mathfrak{n}^2 + \mathfrak{m}B)$, which therefore defines a local homomorphism of A -algebras

$$B \longrightarrow B_1/(\mathfrak{n}_1^2 + \mathfrak{m}B_1) = k[t_1, \dots, t_n]/(t_1, \dots, t_n)^2$$

which can be lifted step by step, by virtue of (iv ter), to homomorphisms of A -algebras from B into $B_1/\mathfrak{n}_1^2$, $B_1/\mathfrak{n}_1^3$, and so on; passing to the projective limit gives the homomorphism $B \rightarrow B_1$ with the desired property.

(v) $\Rightarrow$ (i). As in the commutative diagram

$$\begin{array}{ccccccc} \mathfrak{m}/\mathfrak{m}^2 & \longrightarrow & \mathfrak{n}/\mathfrak{n}^2 & \longrightarrow & \mathfrak{n}/(\mathfrak{n}^2 + \mathfrak{m}B) & \longrightarrow & 0 \\ \downarrow & & \downarrow & & \downarrow & & \\ \mathfrak{m}/\mathfrak{m}^2 & \longrightarrow & \mathfrak{n}_1/\mathfrak{n}_1^2 & \longrightarrow & \mathfrak{n}_1/(\mathfrak{n}_1^2 + \mathfrak{m}B_1) & \longrightarrow & 0 \end{array}$$

the two rows are exact, and the extreme vertical arrows are surjective; the middle arrow is surjective and it follows (since B is complete) that $B \rightarrow B_1$ is *surjective*. Let x_i ($1 \leq i \leq n$)

be elements of B lifting the t_i . They therefore define a homomorphism of A -algebras $B_1 \rightarrow B$, which will be surjective for the same reason as u , and whose composite with u is the identity by construction. Thus $B_1 \rightarrow B$ is also injective, and is consequently an isomorphism. We obtain

COROLLARY 2.3. *Under the conditions of 2.2 (v), u is necessarily an isomorphism.*

This finishes the proof that B is formally smooth over A . At the same time we have recovered 1.4 (though there is little merit in that).

3. Local infinitesimal extension of morphisms into a smooth S -scheme

THEOREM 3.1. *Let $f: X \rightarrow Y$ be a morphism locally of finite type. The following conditions are equivalent:*

- (i) f is smooth.
- (ii) For every prescheme Y' over Y , every closed sub-prescheme Y'_0 of Y' having the same underlying space as Y' , every Y -morphism $g_0: Y'_0 \rightarrow X$, and every $z \in Y'_0$, there exists an open neighborhood U of z in Y' and an extension g of $g_0|_{Y'_0 \cap U}$ to a Y -morphism $U \rightarrow X$.
- (ii bis) For Y' , Y'_0 , and z as in (ii), putting $X' = X \times_Y Y'$, $X'_0 = X \times_Y Y'_0$, every section of X'_0 over Y'_0 extends to a section of X' over an open neighborhood U of z .
- (iii) For every Y -scheme Y' , spectrum of a local artinian ring finite over some $\mathcal{O}_y$ ($y \in Y$), every nonempty closed sub-prescheme Y'_0 of Y' , and every Y -morphism $g_0: Y'_0 \rightarrow X$, there exists a Y -morphism $g: Y' \rightarrow X$ extending g_0 .
- (iii bis) For every Y' , Y'_0 as in (iii), putting $X' = X \times_Y Y'$, $X'_0 = X \times_Y Y'_0$, every section of X'_0 over Y'_0 extends to a section of X' over Y' .

Proof. The equivalence of (ii) and (ii bis), on the one hand, and of (iii) and (iii bis), on the other, is trivial, as is the implication (ii) $\Rightarrow$ (iii). It remains to prove (i) $\Rightarrow$ (ii) and (iii) $\Rightarrow$ (i).

(i) $\Rightarrow$ (ii). Let $x = g_0(z)$. Replacing X by a suitable open neighborhood of x , and Y' by the prescheme induced on the open inverse image of the latter under g_0 , we may suppose that X is étale over $Y[t_1, \dots, t_n]$. Consider the composite Y -morphism $Y'_0 \rightarrow X \rightarrow Y[t_1, \dots, t_n]$; it is defined by n sections of the sheaf $\mathcal{O}_{Y'_0}$, which can therefore be extended in a neighborhood of z to sections of $\mathcal{O}_{Y'}$, so we may suppose that the morphism in question has been extended to a Y -morphism $Y' \rightarrow Y[t_1, \dots, t_n]$. By I.5.6 there is then a unique Y -morphism $g: Y' \rightarrow X$ lifting the preceding one, and at the same time extending g_0 , qed.

(iii) $\Rightarrow$ (i). Since the set of points where f is smooth is open, it suffices to prove that it contains every $x \in X$ that is *closed* in its fiber. Let $y = f(x)$. Then $\mathcal{O}_x$ is an algebra over $\mathcal{O}_y$, a localization of an algebra of finite type, with finite residual extension. On the other hand, hypothesis (iii) implies that every homomorphism from $\mathcal{O}_x$ into an algebra A/J , where A is a local artinian algebra finite over $\mathcal{O}_y$, and J is an ideal contained in its radical, lifts to a homomorphism from $\mathcal{O}_x$ into the algebra A (taking into account that a morphism from $\text{Spec}(B)$, with B local, into X is determined bijectively by a local homomorphism from an $\mathcal{O}_x$ ($x \in X$) into B). By 2.1 it follows that $\mathcal{O}_x$ is formally smooth over $\mathcal{O}_y$, hence smooth over $\mathcal{O}_y$ by 1.7.

COROLLARY 3.2. *Let $f: X \rightarrow Y$ be as in 3.1. The following conditions are equivalent:*

- (i) f is étale.
- (ii) Condition (ii) of 3.1 holds with uniqueness of the extension g of g_0 to U .
- (iii) Condition (iii) of 3.1 holds with uniqueness of g .

It suffices to note, in the proof of (i) $\Rightarrow$ (ii) above, that one can have uniqueness (when Y'_0 is not identical to Y' in a neighborhood of z) only if $n = 0$, a condition that is known to be sufficient.

COROLLARY 3.3. *Let X be a prescheme locally of finite type over a complete local ring A , y the closed point of $Y = \operatorname{Spec}(A)$, and x a point of $f^{-1}(y)$ rational over $k(y)$. If X is smooth over A at x , then there exists a section s of X over Y “passing through x ” i.e. such that $s(y) = x$.*

In particular, if X is smooth over A , then the natural map

$$\Gamma(X/Y) \longrightarrow \Gamma(X \otimes_A k/k)$$

from the sections of X over Y to the set of points of the fiber $f^{-1}(y) = X \otimes_A k$ rational over k , is surjective. This fact was especially well known and used when A is a discrete valuation ring and X is proper over A (in fact, projective over A), in which case the sections of X over Y (i.e. the “points of X with values in A ”) also identify with the rational sections, i.e. with the points of $X \otimes_A K = X_K$ (which is a proper smooth scheme over K) with values in K , the field of fractions of A , i.e. with the points of X rational over K .

4. Local infinitesimal extension of smooth S -schemes

THEOREM 4.1. *Let Y be a locally noetherian prescheme, let Y_0 be a closed subprescheme having the same underlying space, let X_0 be a smooth Y_0 -prescheme, and let x be a point of X_0 . Then there exist an open neighborhood U_0 of x , a prescheme X smooth over Y , and a Y_0 -isomorphism:*

$$h: U_0 \xrightarrow{\sim} X \times_Y Y_0.$$

Moreover, if (U'_0, X', h') is another solution of this problem, then “it is isomorphic to the first in a neighborhood of x ”.

We leave it to the reader to make precise what is meant by this. One may note that for U_0 given, a solution of the stated problem amounts to giving, on U_0 , a sheaf of algebras $\mathcal{B}$ over $f_0^{-1}(\mathcal{O}_Y)$ (where f_0 is the underlying continuous map of the structural morphism $U_0 \rightarrow Y_0$), equipped with a ring homomorphism $\mathcal{B} \rightarrow \mathcal{O}_{U_0}$ compatible with the homomorphism $f_0^{-1}(\mathcal{O}_Y) \rightarrow f_0^{-1}(\mathcal{O}_{Y_0})$, such that

(a) this homomorphism induces an isomorphism

$$\mathcal{B} \otimes_{f_0^{-1}(\mathcal{O}_Y)} f_0^{-1}(\mathcal{O}_{Y_0}) \xrightarrow{\sim} \mathcal{O}_{Y_0};$$

(b) U_0 equipped with $\mathcal{B}$ becomes a smooth Y -prescheme.

In this way, the precise meaning of the assertion of local uniqueness becomes particularly evident.

Proof. We may already suppose that X_0 is étale over some $Y_0[t_1, \dots, t_n] = Y'_0$. But the latter may be regarded as a closed subprescheme of $Y' = Y[t_1, \dots, t_n]$, having the same underlying space. By I 8.3, there exists a prescheme X étale over Y' , and a Y'_0 -isomorphism $X \times_{Y'} Y'_0 \xrightarrow{\sim} X'$. Existence is thus obtained. For uniqueness, one uses property 3.1 (ii) of smooth morphisms, taking into account the following lemma.

LEMMA 4.2. *Let Y be a prescheme, let Y_0 be a closed subprescheme defined by a locally nilpotent sheaf of ideals $\mathcal{J}$, let X and X' be Y -preschemes, and let $u: X \rightarrow X'$ be a Y -morphism. Suppose that X is flat over Y . In order that u be an isomorphism, it is necessary and sufficient that $u_0: X \times_Y Y_0 \rightarrow X' \times_Y Y_0$ be so.*

The proof is easy, by passing to the affine case and looking at the associated graded rings. One should note moreover that the analogous statement, obtained by replacing “isomorphism” by “closed immersion”, is also valid, and without the flatness hypothesis.

REMARK. It is essential to note that the local extension X obtained in 4.1 is *not canonical*, in other words the local isomorphism between two solutions is not unique, i.e. in general there exist nontrivial Y -automorphisms of X inducing the identity on the closed subprescheme $X_0 = X \times_Y Y_0$. This is why, for the construction of *global* infinitesimal extensions of smooth preschemes, one must expect the existence of an obstruction of cohomological nature, which will be made precise below (no. 6).

5. Global infinitesimal extension of morphisms

Let T be a topological space, let $\mathcal{G}$ be a sheaf of groups on X , and let $\mathcal{P}$ be a sheaf of sets on T on which $\mathcal{G}$ acts (on the right, to fix ideas). One says that $\mathcal{P}$ is *formally principal homogeneous* under $\mathcal{G}$ if the familiar homomorphism

$$\mathcal{G} \times \mathcal{P} \longrightarrow \mathcal{P} \times \mathcal{P}$$

of sheaves of sets, deduced from the operations of $\mathcal{G}$ on $\mathcal{P}$, is an *isomorphism*. This amounts to saying that for every $x \in T$, $\mathcal{P}_x$ is *empty or a principal homogeneous space* under the ordinary group $\mathcal{G}_x$, or also that for every open set U of T , $\mathcal{P}(U)$ is empty or a principal homogeneous space under the ordinary group $\mathcal{G}(U)$. One says that $\mathcal{P}$ is a *principal homogeneous sheaf* under $\mathcal{G}$ if it is formally so and if, in addition, the $\mathcal{P}_x$ are nonempty (in other words, if *all* the $\mathcal{P}_x$ are principal homogeneous spaces, hence nonempty, under the $\mathcal{G}_x$)⁴. Recall that the set of classes (up to isomorphism) of principal homogeneous sheaves under $\mathcal{G}$ identifies with the cohomology set $H^1(T, \mathcal{G})$, which is also the usual cohomology group of T with coefficients in $\mathcal{G}$ when $\mathcal{G}$ is commutative. Thus, for every principal homogeneous $\mathcal{P}$, there is a characteristic class $c(\mathcal{P}) \in H^1(T, \mathcal{G})$, whose triviality is necessary and sufficient for $\mathcal{P}$ to be trivial (i.e. isomorphic to $\mathcal{G}$, on which $\mathcal{G}$ acts by right translations), or equivalently for $\mathcal{P}$ to have a section.

PROPOSITION 5.1. *Let S be a prescheme, let X and Y be preschemes over S , and let Y_0 be a closed sub-prescheme of Y defined by an ideal $\mathcal{J}$ on Y of square zero. Let g_0 be an S -morphism from Y_0 to X , and let $\mathcal{P}(g_0)$ be the sheaf on Y whose sections on an open set U are the extensions $g: U \rightarrow X$ of $g_0|_{U \cap Y_0}$ to an S -morphism g . Then $\mathcal{P}(g_0)$ is a sheaf which is naturally formally principal homogeneous under the commutative sheaf of groups*

$$\mathcal{G} = \mathbf{Hom}_{\mathcal{O}_{Y_0}}(g_0^*(\Omega_{X/S}^1), \mathcal{J}).$$

Put $\mathcal{P} = \mathcal{P}(g_0)$. For every open set U of Y we must define a map

$$\mathcal{P}(U) \times \mathcal{G}(U) \longrightarrow \mathcal{P}(U)$$

so that (a), for fixed $g \in \mathcal{P}(U)$, the map $s \mapsto gs$ from $\mathcal{G}(U)$ to $\mathcal{P}(U)$ is bijective, (b) $\mathcal{P}(U)$ becomes a set with group of operators $\mathcal{G}(U)$, and (c) the preceding maps are compatible with the restriction operators for an open set $V \subset U$. The verification of (c) is trivial, so for simplicity one may suppose $U = Y$. The verification of (b) (which is, if one wishes, local in nature) will be left to the reader; we shall limit ourselves, for a fixed $g \in \mathcal{P}(Y)$, to defining a natural bijection from $\mathcal{G}(Y)$ onto $\mathcal{P}(Y)$. Thus suppose that an S -morphism $g: Y \rightarrow X$ is already given, and seek a canonical bijection

$$(*) \quad \mathbf{Hom}_{\mathcal{O}_{Y_0}}(g_0^*(\Omega_{X/S}^1), \mathcal{J}) \xrightarrow{\sim} \mathcal{P}(Y)$$

where $\mathcal{P}(Y)$ is the set of S -morphisms g' from Y to X inducing the same morphism $g_0: Y_0 \rightarrow X$ as g .

The datum of such a g' is equivalent to the datum of an S -morphism $h: Y \rightarrow X \times_S X$ such that $\text{pr}_1 \circ h = g$, and $h \circ i = (g_0, g_0)$, where $\text{pr}_1: X \times_S X \rightarrow X$ is the first projection, $i: Y_0 \rightarrow Y$ is the canonical immersion, and $(g_0, g_0): Y_0 \rightarrow X \times_S X$ is the morphism $\Delta_{X/S} g_0$ with components g_0, g_0 :

$$\begin{array}{ccc} X \times_S X & \xleftarrow{h_0 = (g_0, g_0)} & Y_0 \\ \text{pr}_1 \downarrow & \nwarrow h & \downarrow i \\ X & \xleftarrow{g} & Y \end{array}$$

Since h_0 factors through the diagonal immersion $\Delta_{X/S}$ and since Y is in the first-order infinitesimal neighborhood of Y_0 (i.e. $\mathcal{J}^2 = 0$), the h sought necessarily factor (uniquely)

⁴It seems preferable to adopt the shorter and more expressive term “*torsor under $\mathcal{G}$* ”, introduced in the thesis of J.. GIRAUD.

through the first-order infinitesimal neighborhood of the diagonal. This neighborhood is identified (as an X -prescheme by means of pr_1) with the spectrum X' of the sheaf of algebras $\mathcal{O}_X + \Omega_{X/S}^1$ (where the second term is regarded as a square-zero ideal), the diagonal morphism $X \rightarrow X'$ corresponding to the canonical augmentation of this sheaf of algebras. Put $Y' = X' \times_X Y$, $Y'_0 = Y' \times_Y Y_0 = X' \times_X Y_0$, so that the h sought are in one-to-one correspondence with sections u of Y' over Y extending a given section u_0 of Y'_0 over Y_0 . One may moreover identify Y' with the spectrum of the sheaf of algebras on Y $\mathcal{A} = g^*(\mathcal{O}_X + \Omega_{X/S}^1) = \mathcal{O}_Y + g^*(\Omega_{X/S}^1)$, and Y'_0 with the sheaf of algebras $\mathcal{A}_0 = \mathcal{A} \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y_0} = \mathcal{O}_{Y_0} + g_0^*(\Omega_{X/S}^1)$. Then the section u_0 is the one defined by the canonical augmentation of $\mathcal{A}_0$ into $\mathcal{O}_{Y_0}$. Thus $\mathcal{P}(Y)$ identifies with the set of algebra homomorphisms $\mathcal{A} \rightarrow \mathcal{O}_Y$ inducing the canonical augmentation $\mathcal{A}_0 \rightarrow \mathcal{O}_{Y_0}$. Now the algebra homomorphisms $\mathcal{A} \rightarrow \mathcal{O}_Y$ correspond bijectively to homomorphisms of modules $\mathcal{M} \rightarrow \mathcal{A}$ (putting, for simplicity, $\mathcal{M} = g^*(\Omega_{X/S}^1)$), and we are interested in those which induce the *zero* homomorphism $\mathcal{M}_0 \rightarrow \mathcal{O}_{Y_0}$ (where $\mathcal{M}_0 = \mathcal{M} \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y_0}$), i.e. those which map $\mathcal{M}$ into the ideal $\mathcal{J}$ of the augmentation. We therefore obtain the set $\text{Hom}_{\mathcal{O}_Y}(\mathcal{M}, \mathcal{J}) = \text{Hom}_{\mathcal{O}_{Y_0}}(\mathcal{M}_0, \mathcal{J})$ (since $\mathcal{J}$ is annihilated by $\mathcal{J}$). This is the desired canonical bijection (*).

Taking into account the implication (i) $\Rightarrow$ (iii) in 3.1, one obtains:

COROLLARY 5.2. *If X is smooth over S (at least at the points of $g_0(Y_0)$), then $\mathcal{P}$ is even a principal homogeneous sheaf under the commutative sheaf of groups $\mathcal{G}$, which in this case may also be written*

$$\mathcal{G} = g_0^*(\mathfrak{g}_{X/S}) \otimes_{\mathcal{O}_{Y_0}} \mathcal{J}$$

where $\mathfrak{g}_{X/S}$ is the sheaf on X dual to $\Omega_{X/S}^1$, i.e. the tangent sheaf (or sheaf of derivations) of X relative to S . (This last formula follows from the fact that $\Omega_{X/S}^1$ is then locally free of finite type.)

In particular, this principal homogeneous sheaf determines a cohomology class in $H^1(Y_0, \mathcal{G})$, whose vanishing is necessary and sufficient for the existence of an S -morphism g extending g_0 . And if such an extension exists, the set of all possible extensions is a homogeneous space under the group $H^0(Y_0, \mathcal{G})$.

In applying the methods of formal geometry, the situation is most often the following: two S -preschemes X and Y are given, together with a coherent ideal $\mathcal{I}$ on S ; let S_n be the closed sub-prescheme of S defined by $\mathcal{I}^{n+1}$, and put

$$X_n = X \times_S S_n, \quad Y_n = Y \times_S S_n.$$

Suppose that one has an S_n -morphism

$$g_n: Y_n \longrightarrow X_n$$

(or, what amounts to the same thing, an S -morphism $Y_n \rightarrow X$, or again an S_{n+1} -morphism $Y_n \rightarrow X_{n+1}$, since such a morphism necessarily induces $Y_n \rightarrow X_n$), and one seeks to extend it to an S_{n+1} -morphism

$$g_{n+1}: Y_{n+1} \longrightarrow X_{n+1}.$$

(If one can continue indefinitely, one thus obtains a morphism $\hat{Y} \rightarrow \hat{X}$ for the formal preschemes completed from Y and X for the ideals $\mathcal{I}\mathcal{O}_Y$ and $\mathcal{I}\mathcal{O}_X$.) One may apply 5.1 after replacing (S, X, Y, Y_0, g_0) by $(S_{n+1}, X_{n+1}, Y_{n+1}, Y_n, g_n)$. The sheaf $\mathcal{G}$ becomes here the sheaf of homomorphisms of modules from $g_n^*(\Omega_{X_{n+1}/S_{n+1}}^1)$ to $\mathcal{J} = \mathcal{I}^{n+1}\mathcal{O}_Y/\mathcal{I}^{n+2}\mathcal{O}_Y$. Since $\mathcal{J}$ is annihilated by $\mathcal{I}\mathcal{O}_Y$, one can replace $g_n^*(\Omega_{X_{n+1}/S_{n+1}}^1)$ by the sheaf it induces on Y_0 , namely $h_0^*(\Omega_{X/S}^1)$, where h_0 is the composite $Y_0 \rightarrow Y_n \rightarrow X_{n+1}$, or again the composite $Y_0 \rightarrow X_0 \rightarrow X_{n+1}$, where $g_0: Y_0 \rightarrow X_0$ is induced by g_n . Since the inverse image of $\Omega_{X_{n+1}/S_{n+1}}^1$ on $X_0 = X_{n+1} \times_{S_{n+1}} S_0$ is Ω_{X_0/S_0}^1 , one sees that one also has

$$\mathcal{G} = \mathbf{Hom}_{\mathcal{O}_{Y_0}}(g_0^*(\Omega_{X_0/S_0}^1), \mathcal{I}^{n+1}\mathcal{O}_Y/\mathcal{I}^{n+2}\mathcal{O}_Y).$$

Thus one obtains the

COROLLARY 5.3. *Let $S, X, Y, \mathcal{I}, g_n$ be as above, and let $\mathcal{P}(g_n)$ be the sheaf on Y whose sections on an open set U are the extensions g_{n+1} of g_n to an S_{n+1} -morphism $Y_{n+1} \rightarrow X_{n+1}$. Then $\mathcal{P}(g_n)$ is a formally principal homogeneous sheaf under the sheaf of groups*

$$\mathcal{G} = \mathbf{Hom}_{\mathcal{O}_{Y_0}}(g_0^*(\Omega_{X_0/S_0}^1), \mathrm{gr}_{\mathcal{I}\mathcal{O}_Y}^{n+1}(\mathcal{O}_Y)).$$

In particular:

COROLLARY 5.4. *If in addition X is smooth over S (at least at the points of $g_0(Y_0)$), then $\mathcal{P}(g_n)$ is even a principal homogeneous sheaf. In particular, it defines an obstruction class in $H^1(Y_0, \mathcal{G})$, whose vanishing is necessary and sufficient for the existence of a global extension g_{n+1} of g_n . And if such an extension exists, the set of all global extensions is a principal homogeneous space under $H^0(Y_0, \mathcal{G})$. Finally, in the case under consideration, the sheaf $\mathcal{G}$ may also be written*

$$\mathcal{G} = g_0^*(\mathfrak{g}_{X_0/S_0}) \otimes_{\mathcal{O}_{Y_0}} \mathrm{gr}_{\mathcal{I}\mathcal{O}_Y}^{n+1}(\mathcal{O}_Y)$$

Proceeding step by step, one sees therefore that if all the $H^1(Y_0, \mathcal{G}_n)$ vanish (where $\mathcal{G}_n = g_0^*(\mathfrak{g}_{X_0/S_0}) \otimes \mathrm{gr}_{\mathcal{I}\mathcal{O}_Y}^n(\mathcal{O}_Y)$), then, starting with an arbitrary g_k , one can extend it successively to $g_{k+1}, \dots$. In particular, if $\mathcal{I}$ is nilpotent, one will be able to find an extension g of g_k to Y . The condition that the H^1 vanish is satisfied in particular if Y_0 is affine. Thus one obtains:

COROLLARY 5.5. *In the statement of the theorem 3.1, one obtains a necessary and sufficient condition equivalent to the others by supposing that the Y' which occurs in (ii) (or (ii bis)) is affine, and by requiring the existence of a global extension g of g_0 to all of Y' .*

It should be noted that the proof of 3.1 could not give us this result directly.

An important case is that in which Y is flat over S ; then one has

$$\mathrm{gr}^n(\mathcal{O}_Y) = \mathrm{gr}^n(\mathcal{O}_S) \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{Y_0}$$

and when, in addition, the $\mathrm{gr}^n(\mathcal{O}_S)$ are locally free on S , one obtains

$$\mathcal{G}_n = \mathbf{Hom}_{\mathcal{O}_{Y_0}}(g_0^*(\Omega_{X_0/S_0}^1), \mathcal{O}_{Y_0}) \otimes_{\mathcal{O}_{S_0}} \mathrm{gr}^n(\mathcal{O}_S),$$

or again, if Ω_{X_0/S_0}^1 is itself locally free (for example, if X is smooth over S),

$$\mathcal{G}_n = g_0^*(\mathfrak{g}_{X_0/S_0}) \otimes_{\mathcal{O}_{S_0}} \mathrm{gr}^n(\mathcal{O}_S)$$

If, for example, S is affine with affine ring A , and $\mathcal{I}$ is defined by an ideal I of A , one obtains

$$H^i(Y_0, \mathcal{G}_n) = H^i(Y_0, \mathcal{G}_0) \otimes_A \mathrm{gr}_I^n(A)$$

for every i (indeed, the question is local on S_0 , and one is reduced to the case where one tensors by a free module). In this case, the vanishing of $H^1(Y_0, \mathcal{G}_0)$ implies that all the obstructions to the successive extensions of g_n vanish. Thus one obtains:

COROLLARY 5.6. *Let $(S, X, Y, \mathcal{I}, g_n)$ be as above, suppose in addition that X is smooth over S and Y is flat over S , and finally that S is affine, and that the $\mathrm{gr}^n(\mathcal{O}_S) = \mathcal{I}^n/\mathcal{I}^{n+1}$ are locally free. Then the obstruction to constructing g_{n+1} lies in $H^1(Y_0, \mathcal{G}_0) \otimes_A \mathrm{gr}_I^{n+1}(A)$ (where A is the ring of S , I the ideal of A defining $\mathcal{I}$), on putting*

$$\mathcal{G}_0 = g_0^*(\mathfrak{g}_{X_0/S_0}).$$

If $H^1(Y_0, \mathcal{G}_0) = 0$, then g_n can be extended to an $\widehat{S}$ -morphism $\widehat{g}: \widehat{Y} \rightarrow \widehat{X}$.

Of course, this result would remain valid exactly as stated if, instead of starting with ordinary S -preschemes X and Y , one started with formal $\widehat{\mathcal{I}}$ -adic $\widehat{S}$ -preschemes $\mathfrak{X}$ and $\mathfrak{Y}$. It allows one to prove, for example, that certain formal schemes proper over a complete local ring are in fact algebraic. Indeed, proceeding as in the lemma 4.2, one obtains:

COROLLARY 5.7. *Under the conditions of 5.6, if g_0 is an isomorphism, then so is $\widehat{g}$.*

(N.B. the same result holds for closed immersions.)

Thus one obtains:

PROPOSITION 5.8. *Let A be a complete local ring with maximal ideal $\mathfrak{m}$ and residue field k , and let $\mathfrak{X}$ and $\mathfrak{Y}$ be two $\mathfrak{m}$ -adic formal preschemes over A , flat over A (i.e. for every n , X_n and Y_n are flat over $A_n = A/\mathfrak{m}^{n+1}$). Suppose that $X_0 = \mathfrak{X} \otimes_A k$ is smooth over k and that $H^1(X_0, \mathfrak{g}_{X_0/k}) = 0$. Then every k -isomorphism from Y_0 onto X_0 extends to an A -isomorphism from $\mathfrak{Y}$ onto $\mathfrak{X}$; this extension is unique if, in addition, $H^0(X_0, \mathfrak{g}_{X_0/k}) = 0$.*

This gives in particular a result on the *uniqueness* of a formal prescheme smooth over A reducing to a given prescheme X_0 (provided $H^1(X_0, \mathfrak{g}_{X_0/k}) = 0$). Moreover, if $\mathfrak{X}$ and $\mathfrak{Y}$ come from ordinary proper schemes over A , say X and Y , then one knows, by the existence theorem for sheaves in formal geometry (cf. the exposé of the Bourbaki Seminar, no. 182⁵), that there is a one-to-one correspondence between the A -isomorphisms $Y \xrightarrow{\sim} X$ and the A -isomorphisms of the formal completions, hence

COROLLARY 5.9. *The preceding statement 5.8 remains valid on replacing $\mathfrak{X}$ and $\mathfrak{Y}$ by ordinary A -schemes X and Y , proper over A .*

Finally, when $\mathfrak{X}$ is a formal scheme proper over A , and $\mathfrak{Y}$ is of the form $\widehat{Y}$ where Y is an ordinary proper scheme over A , Proposition 5.8 gives sufficient conditions for finding an isomorphism of $\mathfrak{X}$ with $\widehat{Y}$, hence for the formal scheme $\mathfrak{X}$ to be in fact “algebraic” (i.e. isomorphic to a $\widehat{X}$, X an ordinary proper scheme over A , which will then be canonically determined). This is notably what happens if $X_0 = \mathbf{P}_k^r$ (or more generally, if X_0 is a Severi–Brauer scheme, i.e. becomes isomorphic to the standard projective space over the algebraic closure of k): every formal scheme proper and flat over A , with fiber $\mathbf{P}_k^r$, is algebraizable, and more precisely is isomorphic to the $\mathfrak{m}$ -adic formal completion of $\mathbf{P}_A^r$. In particular (thanks to the “existence theorem”) every ordinary proper scheme over A , with fiber $\mathbf{P}_k^r$, is isomorphic to $\mathbf{P}_A^r$ (A being a complete local ring). Using descent theory, one can prove that if A is not complete, X becomes isomorphic to $\mathbf{P}^r$ after making a finite étale extension $A \rightarrow A'$ of the base (and in this form, the result remains valid for a fiber which is a Severi–Brauer scheme).

6. Global infinitesimal extension of smooth S -schemes

Under the conditions of Theorem 4.1, we propose to seek whether there exists a prescheme X smooth over Y such that $X \times_Y Y_0$ is Y_0 -isomorphic to X_0 , knowing that such a scheme “exists locally on X_0 .” Taking up again the method of construction step by step, one is led to replace Y by the letter S , to suppose that one is given a closed sub-prescheme S_0 of S defined by a sheaf of ideals $\mathcal{I}$ (which it is no longer necessary to suppose locally nilpotent), to introduce the closed sub-preschemes S_n of S defined by $\mathcal{I}^{n+1}$, and to suppose that one is given a sub-prescheme X_n smooth over S_n . One proposes to find an S_{n+1} -prescheme X_{n+1} “reducing to X_n ,” i.e. equipped with an isomorphism

$$X_{n+1} \times_{S_{n+1}} S_n \xleftarrow{\sim} X_n$$

which is *smooth* over S_{n+1} (or, what amounts to the same thing by II 2.1, *flat* over S_{n+1}). As we noted in 4, such data amounts to giving a sheaf of algebras $\mathcal{B}$ on $f^{-1}(\mathcal{O}_{S_{n+1}})$ (where f is the underlying continuous map of the structural morphism $X_n \rightarrow S_n$), equipped with an augmentation $\mathcal{B} \rightarrow \mathcal{O}_{X_n}$ compatible with the augmentation $f^{-1}(\mathcal{O}_{S_{n+1}}) \rightarrow f^{-1}(\mathcal{O}_{S_n})$, and satisfying two conditions (a) and (b) that we shall not rewrite, merely noting that they are of a *local nature* on the topological space underlying X_n . By 4.1 one knows that a solution exists locally. It is moreover unique up to (nonunique) isomorphism, at least locally. Let us begin by making this point precise:

⁵Cf. EGA III 5.4.1 for the proof.

PROPOSITION 6.1. *Let X_{n+1} over S_{n+1} reduce to X_n over S_n . Then the sheaf (on the topological space underlying X_n , or equivalently X_0) of the S_{n+1} -automorphisms of X_{n+1} which induce the identity on X_n is canonically isomorphic to*

$$\mathcal{G} = \mathfrak{g}_{X_0/S_0} \otimes_{\mathcal{O}_{S_0}} gr_{\mathcal{I}}^{n+1}(\mathcal{O}_S)$$

(as a sheaf of groups).

Indeed, by 5.4 and 4.2 this sheaf is a principal homogeneous sheaf under $\mathcal{G}$. Since it is equipped with a distinguished section (the identity automorphism of X_{n+1}), it is therefore identified, as a sheaf of sets, with $\mathcal{G}$. One must verify that this identification is compatible with the group structures. This is easy, and is moreover a special case of a more general result on the compatibility of the principal-bundle structures in 5.1 and 5.3 with composition of morphisms (a result that we do not state here, but which ought to appear in the hyperplodocus).

In particular, the sheaf on X_0 of germs of automorphisms of X_{n+1} (with the structures just made explicit) is *commutative*. It follows that if X'_{n+1} is another solution of the problem, isomorphic to X_{n+1} over the open U of X_0 , then the isomorphism from $\mathbf{Aut}(X_{n+1})|U$ to $\mathbf{Aut}(X'_{n+1})|U$ deduced by transport of structure from an isomorphism $X_{n+1}|U \xrightarrow{\sim} X'_{n+1}|U$ does not depend on the choice of the latter. (It is in fact nothing other than the identity isomorphism of $\mathcal{G}$, when the two automorphism sheaves are identified with $\mathcal{G}$ by 6.1.)

From 6.1 one deduces:

COROLLARY 6.2. *Let X_{n+1} and X'_{n+1} be smooth over S_{n+1} and “reducing to X_n .” Then the sheaf (on the space underlying X_0) of S_{n+1} -isomorphisms from X_{n+1} to X'_{n+1} inducing the identity on X_n is naturally a principal homogeneous sheaf under $\mathcal{G}$.*

This indeed expresses that X_{n+1} and X'_{n+1} are locally isomorphic, and that the sheaf of germs of automorphisms of the first is $\mathcal{G}$.

Now note that by 4.1 one can always find a covering (U_i) of X_n by open sets (which may be supposed affine), and for every i a smooth scheme X^i over S_{n+1} reducing to U_i . Suppose for simplicity that X_n is *separated*, so that the $U_{ij} = U_i \cap U_j$ are still *affine* open subsets of X_n . Since the H^1 of such an open set, with values in the quasi-coherent sheaf $\mathcal{G}$, is zero, it follows by the corollary 6.2 that $X^i|U_{ij}$ is isomorphic to $X^j|U_{ij}$; let

$$f_{ji} : X^i|U_{ij} \xrightarrow{\sim} X^j|U_{ij}$$

be such an isomorphism. It is determined up to a section of $\mathcal{G}$ on U_{ij} . Put, for every triplet of indices,

$$f_{ji}^{(k)} = f_{ji}|U_{ijk} \quad \text{where } U_{ijk} = U_i \cap U_j \cap U_k.$$

If one had

$$(1) \quad f_{kj}^{(i)} f_{ji}^{(k)} = f_{ki}^{(j)},$$

it would follow that the X^i glue by the f_{ji} , and hence define a solution $X = X_{n+1}$ of the problem sought. Such a solution exists more generally if one can modify the f_{ji} into f'_{ji} :

$$(2) \quad f'_{ji} = f_{ji} g_{ji} \quad (g_{ji} \in \Gamma(U_{ij}, \mathcal{G}))$$

so that the f'_{ji} satisfy the transitivity condition above. This sufficient condition for the existence of a solution is also necessary, as one sees by recalling that such a solution X must, on each U_i , be isomorphic to X^i , which therefore allows one to choose isomorphisms

$$f_i : X|U_i \xrightarrow{\sim} X^i$$

and to define

$$f'_{ji} = (f_j|U_{ij})(f_i|U_{ij})^{-1} : X^i|U_{ij} \xrightarrow{\sim} X^j|U_{ij}$$

satisfying the gluing condition.

Now put

$$(3) \quad f_{ijk} = (f_{ki}^{(j)})^{-1} f_{kj}^{(i)} f_{ji}^{(k)},$$

it is an automorphism of $X^i|U_{ijk}$, which we identify with a section of $\mathcal{G}$ by 6.1. One observes that it is a 2-cocycle f of the open covering $\mathcal{U} = (U_i)$, with coefficients in $\mathcal{G}$, by a small formal calculation left to the reader. The same calculation shows that, subject to (2), the gluing condition (1) for the f'_{ij} is equivalent to the formula

$$(4) \quad f = dg,$$

where $g = (g_{ij})$ is regarded as a 1-cochain of $\mathcal{U}$ with coefficients in $\mathcal{G}$. Thus *the necessary and sufficient condition for the existence of a solution of the problem is that the cohomology class in $H^2(\mathcal{U}, \mathcal{G})$ defined by the cocycle (3) be zero*. Note moreover that since $\mathcal{U} = (U_i)$ is an affine covering of X_0 , which is a *scheme*, $H^2(\mathcal{U}, \mathcal{G})$ identifies with $H^2(X_0, \mathcal{G})$. It is immediate that the cohomology class thus obtained in $H^2(X_0, \mathcal{G})$ does not depend on the affine covering considered. We shall call it the *obstruction class to extending X_n to a scheme X_{n+1} smooth over S_{n+1}* .

Suppose this obstruction is zero. Then the argument sketched above shows that every solution $X = X_{n+1}$ is isomorphic to a solution obtained by gluing from isomorphisms f'_{ji} , which may be supposed to have the form (2), the gluing condition being none other than (3). The set of admissible g is therefore a principal homogeneous space under the group $Z^1(\mathcal{U}, \mathcal{G})$ of 1-cocycles of $\mathcal{U}$ with coefficients in $\mathcal{G}$. Moreover, one sees at once that *two cochains g and g' (such that $dg = dg' = f$) define isomorphic solutions if and only if the cocycle $g - g'$ is of the form dh , where $h = (h_i) \in C^0(\mathcal{U}, \mathcal{G})$* . Thus one obtains:

THEOREM 6.3. *Let $(S, \mathcal{I}, X_n)$ be as above, with X_n assumed separated⁶. Then one can define canonically an obstruction class in $H^2(X_0, \mathcal{G})$ (where $\mathcal{G}$ is defined in 6.1), whose vanishing is necessary and sufficient for the existence of a scheme X_{n+1} smooth over S_{n+1} reducing to X_n . If this obstruction is zero, then the set of isomorphism classes, up to isomorphism (inducing the identity on X_n), of S_{n+1} -preschemes X_{n+1} reducing to X_n is naturally a principal homogeneous space under $H^1(X_0, \mathcal{G})$.*

REMARKS. Starting from 6.1, the arguments made here are purely formal, and are advantageously transcribed in the setting of local categories, or even of general fibered categories. The obstruction class to the existence of a “global” object of a category (in which one can find an object “locally,” and in which two objects are always “locally isomorphic,” the automorphism group of every object being commutative) obtained in this general context contains as a special case the “second boundary homomorphism” in an exact sequence of sheaves of not necessarily commutative groups, studied for example by Grothendieck in Kansas or Tôhoku. The silly cocycle calculation made here should therefore be regarded as a makeshift, due to the absence of a satisfactory reference text.

6.1. It should be noted that in 6.3 there is in general no distinguished element in the principal homogeneous space under $H^1(X_0, \mathcal{G})$ under consideration. This is reflected in particular by the fact that, after localizing on S , one obtains a principal homogeneous sheaf on S_0 , with structural group $\mathbf{R}^1 f_*(\mathcal{G})$, which is not necessarily trivial, i.e. which defines a cohomology class in $H^1(S_0, \mathbf{R}^1 f_*(\mathcal{G}))$ which is not necessarily zero. (This is when one supposes that the class $d \in H^2(X_0, \mathcal{G})$ is not zero, but is zero “locally over S ,” i.e. defines a zero section of $\mathbf{R}^2 f_*(\mathcal{G})$, i.e. a zero element in $H^0(S_0, \mathbf{R}^2 f_*(\mathcal{G}))$.)

6.2. For the moment one knows almost nothing about the general algebraic mechanism of the cohomology classes introduced in this number and their relations with the preceding number, and one knows nothing precise to say about them in the simplest particular cases, such as the case of abelian schemes over artinian rings⁷. One hopes that there will be people

⁶This condition is in fact unnecessary, and one can avoid the cocycle calculations above. Cf. the book by J. GIRAUD, *Cohomologie Non Abélienne* (forthcoming from Springer Verlag, 1971). Compare Remarks 6.

⁷It is now known that this obstruction is always zero in this case [added in 2003 (MR): cf. F. Oort, *Finite group schemes, local moduli for abelian varieties and lifting problems, Algebraic Geometry Oslo 1970*, Wolters-Noordhoff, 1972, pp. 223–254].

to work the question out thoroughly; it seems particularly interesting. It is intimately linked, in particular, to the “module theory” of algebraic structures.

COROLLARY 6.4. *Suppose that $H^2(X_0, \mathcal{G}) = 0$. Then an X_{n+1} exists, and it is unique up to isomorphism if moreover $H^1(X_0, \mathcal{G}) = 0$.*

In particular, one concludes, proceeding step by step (and observing that an affine scheme is acyclic for a quasi-coherent sheaf):

COROLLARY 6.5. *Under the conditions of Theorem 4.1, if X_0 is affine, then there exists an X smooth over Y reducing to X_0 , and this X is unique up to (nonunique) isomorphism.*

It should be noted that the direct proof of Theorem 4.1 could not have given this result.

COROLLARY 6.6. *Under the conditions of 6.3, suppose S affine with ring A , $\mathcal{I}$ defined by an ideal I of A , and finally that the $\mathrm{gr}_{\mathcal{I}}^n(\mathcal{O}_S) = \mathcal{I}^n/\mathcal{I}^{n+1}$ are locally free. Then $H^i(X_0, \mathcal{G})$ identifies with*

$$H^i(X_0, \mathcal{G}_0) \otimes_A \mathrm{gr}_I^{n+1}(A),$$

where

$$\mathcal{G}_0 = \mathfrak{g}_{X_0/S_0},$$

so the obstruction class to extending X_n lies in $H^2(X_0, \mathcal{G}_0) \otimes_A \mathrm{gr}_I^{n+1}(A)$, and, if it is zero, the set of isomorphism classes (up to isomorphism) of solutions is a principal homogeneous space under $H^1(X_0, \mathcal{G}_0) \otimes_A \mathrm{gr}_I^{n+1}(A)$.

In particular:

COROLLARY 6.7. *Under the conditions of 6.6, suppose*

$$H^2(X_0, \mathfrak{g}_{X_0/S_0}) = 0,$$

then there exists an $\hat{I}$ -adic formal scheme $\mathfrak{X}$ over the $\mathcal{I}$ -adic formal completion $\hat{S}$ of S , which is “smooth over S ” (i.e. the $\mathfrak{X}_p$ are smooth over the S_p) and which reduces to X_n , i.e. is equipped with an isomorphism

$$\mathfrak{X} \times_S S_n \xrightarrow{\sim} X_n.$$

If moreover $H^1(X_0, \mathfrak{g}_{X_0/S_0}) = 0$, then such a $\mathfrak{X}$ is unique up to isomorphism.

Indeed, one constructs X_{n+1} , X_{n+2} , and so on, step by step, whence $\mathfrak{X}$ by passing to the inductive limit of the X_i . The assertion of uniqueness already appears in the preceding number.

7. Application to the construction of formal schemes and smooth ordinary schemes over a complete local ring A

The results of the preceding no. sometimes make it possible to prove the existence of an $\mathfrak{m}$ -adic formal scheme $\mathfrak{X}$ over such a ring, reducing to a given smooth scheme X_0 over k . Let us distinguish two cases:

a) A is “of equal characteristic” (this is the case in particular if k has characteristic 0). Then one knows that there exists a subfield of representatives of A , i.e. a subfield k' such that $A \rightarrow k$ induces an isomorphism $k' \xrightarrow{\sim} k$. There even exists a smooth ordinary scheme over A reducing to X_0 , namely $X = X_0 \otimes_k A$, with A regarded as an algebra over k by means of the homomorphism $k \rightarrow k' \rightarrow A$ defined by k' . It should nevertheless be noted that this construction is not “natural”; it is easy to convince oneself (already in the case where $A = k[t]/(t^2)$, the algebra of dual numbers) that another lifting homomorphism $k \rightarrow A$ (defined in this instance by an absolute derivation of k into itself) defines an X' over A which in general is not isomorphic to X (if $H^1(X_0, \mathfrak{g}_{X_0/k}) \neq 0$). It would moreover be interesting to study (for k of characteristic 0, or non-perfect and of characteristic $p > 0$) which smooth X over A are obtained in this way, and under what condition two homomorphisms $k \rightarrow A$ define isomorphic A -schemes. Nevertheless, the existence of k' suffices to entail that the first

obstruction to lifting X_0 , which is in $H^2(X_0, \mathfrak{g}_{X_0/k}) \otimes_k \mathfrak{m}/\mathfrak{m}^2$, is necessarily zero. Of course, when one has then lifted X_0 to X_1 smooth over $A/\mathfrak{m}^2$, the new obstruction to constructing X_2 will in general not be zero: it will be a function of a variable element in a certain principal homogeneous space under $H^1(X_0, \mathcal{G}_0) \otimes \mathfrak{m}/\mathfrak{m}^2$ and lies in $H^2(X_0, \mathcal{G}_0) \otimes \mathfrak{m}^2/\mathfrak{m}^3$: it would be appropriate to study the situation in detail⁸.

b) *A is of unequal characteristics.* In this case, we know nothing (except if by chance $H^2(X_0, \mathfrak{g}_{X_0/k}) = 0$, in which case one can construct a simple $\mathfrak{m}$ -adic formal scheme over A reducing to k). Even if $A = \mathbf{Z}/p^2\mathbf{Z}$ and X_0 is an “abelian” scheme of dimension 2, one does not know whether it can be lifted to an $X = X_1$ smooth over A^9 , on the other hand, one has no example of an X_0 which one has proved does not come from an ordinary scheme X smooth over A . (I have the impression that such an example must exist, with X_0 a projective surface)¹⁰. Let us simply point out that, according to Cohen’s theorem, there exists a Cohen p -ring B with residue field k and a homomorphism $B \rightarrow A$ inducing the identity isomorphism on the residue fields; consequently, the “strongest” lifting result would be obtained by taking A a Cohen p -ring: if there exists a solution (ordinary or formal) over such a ring, there exists one over every complete local ring with residue field k . In particular, since for a Cohen p -ring $\mathfrak{m}/\mathfrak{m}^2$ is canonically identified with k , one sees that *for every smooth scheme X_0 over a field k of characteristic $p > 0$, there exists a cohomology class in $H^2(X_0, \mathfrak{g}_{X_0/k})$ which is the first obstruction to lifting X_0 to a smooth scheme over a Cohen p -ring; one does not know whether it can be nonzero*¹¹.

Even if one succeeds step by step in constructing the X_n reducing to X_0 , this generally gives only a *formal* scheme $\mathfrak{X}$ smooth over A , reducing to X_0 . When X_0 is proper over A , the question remains whether $\mathfrak{X}$ is in fact algebraizable, in order to obtain an *ordinary* scheme proper over A and simple over A , reducing to X_0 . The only known criterion (mentioned in the Bourbaki Seminar, and appearing in the Elements, Chapter III, 4.7.1) is the following: if $\mathfrak{X}$ is proper over A , and if $\mathcal{L}$ is an invertible sheaf on $\mathfrak{X}$ such that the sheaf induced $\mathcal{L}_0$ on X_0 is ample (i.e. a suitable tensor power $\mathcal{L}_0^{\otimes n}$, $n > 0$, comes from a projective embedding of X_0), then there exists a scheme X projective over A , and an ample invertible sheaf on X , such that $(\mathfrak{X}, \mathcal{L})$ is obtained from it by $\mathfrak{m}$ -adic completion. This therefore leads us, given a locally free sheaf $\mathcal{E}_0$ on X_0 (which we shall choose ample and invertible for our purpose), to extend it to a locally free sheaf $\mathcal{E}$ on $\mathfrak{X}$. For this, we are reduced to constructing step by step locally free sheaves $\mathcal{E}_n$ on the X_n . The discussion is entirely analogous to that in no. 6, (cf. remark 6), the essential role being played by the *sheaf of automorphisms* of an $\mathcal{E}_{n+1}$ which induce the identity on $\mathcal{E}_n$: one immediately shows that this sheaf is identified with

$$(*) \quad \mathcal{G} = \mathbf{Hom}_{\mathcal{O}_{X_0}}(\mathcal{E}_0, \mathcal{E}_0 \otimes \mathrm{gr}_{\mathcal{I}}^{n+1}(\mathcal{O}_X)) = \mathbf{Hom}_{\mathcal{O}_{X_0}}(\mathcal{E}_0, \mathcal{E}_0) \otimes \mathrm{gr}_{\mathcal{I}}^{n+1}(\mathcal{O}_X)$$

which is again a sheaf of commutative groups. We find:

PROPOSITION 7.1. *Let S be a prescheme endowed with a quasi-coherent sheaf of ideals $\mathcal{I}$, X a prescheme over S , S_n the subprescheme of S defined by $\mathcal{I}^{n+1}$, and $X_n = X \times_S S_n$ (for every integer n). Let $\mathcal{E}_n$ be a locally free sheaf on X_n ; we propose to extend it to a locally free sheaf $\mathcal{E}_{n+1}$ on X_{n+1} . Then $\mathcal{E}_n$ defines a canonical obstruction class in $H^2(X_0, \mathcal{G})$, where $\mathcal{G}$ is the quasi-coherent sheaf given by the formula above, whose vanishing is necessary and sufficient for the existence of an $\mathcal{E}_{n+1}$ extending $\mathcal{E}_n$. If this class is zero, then the set of classes, up to isomorphism (inducing the identity on $\mathcal{E}_n$), of solutions $\mathcal{E}_{n+1}$, is a principal homogeneous space under $H^1(X_0, \mathcal{G})$.*

⁸It is no doubt described by the Kodaira–Spencer bracket operation (cf. Cartan Seminar, 1960/61, Exp.. 4).

⁹This has now been proved, cf. note 7 page 13.

¹⁰Such an example has since been constructed by J-P. Serre (Proc. Nat. Acad. Sc. USA **47** (1961), no. 1, p.. 108–109), at least for certain dimensions. D.. Mumford gave an example (unpublished) with an *algebraic* surface.

¹¹It can be nonzero, as indicated in note 10 on page 15.

This proposition gives rise to the usual corollaries. Let us point out only that if X is *flat* over S , then one can write

$$\mathcal{G} = \mathbf{Hom}_{\mathcal{O}_{X_0}}(\mathcal{E}_0, \mathcal{E}_0) \otimes_{\mathcal{O}_{S_0}} \mathrm{gr}_{\mathcal{I}}^{n+1}(\mathcal{O}_S)$$

whence, if S is affine of ring A and if the $\mathcal{I}^n/\mathcal{I}^{n+1}$ are locally free, the sufficient condition

$$H^2(X_0, \mathcal{G}_0) = 0 \quad \text{with} \quad \mathcal{G}_0 = \mathbf{Hom}_{\mathcal{O}_{X_0}}(\mathcal{E}_0, \mathcal{E}_0)$$

for the existence of an $\mathcal{E}_{n+1}$, and hence step by step for the existence of successive extensions $\mathcal{E}_m$ ($m = n, n+1$, etc.).

Returning to the initial situation, we therefore find:

PROPOSITION 7.2. *Let A be a complete local ring, $\mathfrak{X}$ a proper and flat formal scheme over A , such that X_0 is projective and $H^2(X_0, \mathcal{O}_{X_0}) = 0$. Then there exists a scheme X projective over A whose $\mathfrak{m}$ -adic formal completion is isomorphic to $\mathfrak{X}$.*

Combining this with 6.7, we find:

THEOREM 7.3. *Let A be a complete local ring with residue field k , X_0 a projective and smooth scheme over k , such that*

$$H^2(X_0, \mathfrak{g}_{X_0/k}) = H^2(X_0, \mathcal{O}_{X_0}) = 0$$

Then there exists a smooth and projective scheme X over A , reducing to X_0 .

More generally, if one is given an X_n smooth over $A_n = A/\mathfrak{m}^{n+1}$ reducing to X_0 , then there exists an X smooth and proper over A and an isomorphism $X \otimes_A A_n = X_n$.

COROLLARY 7.4. *Every smooth and proper curve over k is obtained by reduction from a smooth and proper curve over A .*

It is this result which will be the essential tool (together with the existence theorem for sheaves in Formal Geometry) for studying the fundamental group of X_0 by transcendental means.